Второй семестр

Первое задание

№ 1.

$V = const;\ T_1;\ T_2.$ $\quad \Delta U - ?$ $\quad U = \frac{i}{2}\nu RT \Rightarrow \Delta U = \frac{iR}{2}(\nu_2 T_2 - \nu_1 T_1)$

$pV = \nu RT,\quad p, V = const \Rightarrow \nu T = const \Rightarrow \Delta U = 0.$

Ответ: $\Delta U = 0.$

№ 2.

$T = const;\ V_0 \to 2V_0 \Rightarrow \delta A = p\,dV = \nu RT \frac{dV}{V} \Rightarrow$

$\Rightarrow \Delta A = \nu RT \int_{V_0}^{2V_0} \frac{dV}{V} = \nu RT \ln\frac{2V_0}{V_0} = \nu RT \ln 2 \Rightarrow$

$\Rightarrow \Delta A \approx 1{,}7 \cdot 10^3$ Дж.

№ 3.

$T = 273K;\ \Delta T = 1K.$ $\quad \Delta C_s - ?$ $\quad C_s = \sqrt{\gamma \frac{RT}{\mu}} \Rightarrow \Delta C_s = \frac{\partial C_s}{\partial T}\Delta T =$

$= \frac{1}{2}\sqrt{\gamma\frac{R}{\mu}}\,\frac{\Delta T}{\sqrt{T}} = \frac{1}{2}\sqrt{\frac{\gamma RT}{\mu}}\,\frac{\Delta T}{T} = \frac{C_s \Delta T}{2T} \Rightarrow \Delta C_s \approx 0{,}61$ м/с.

№ 1.40.

$V_0 = 4$ л; $V_1 = 1$ л; $p_0 = 1$ атм; $p_1 = 8$ атм; $T_0 = 300$ K. C-?

$C = const \Rightarrow C\,dT = C_V\,dT + \frac{\nu RT}{V}dV \Rightarrow$

$\Rightarrow (C - C_V)\frac{dT}{T} = \nu R \frac{dV}{V} \Rightarrow$

$\Rightarrow (C - C_V)\ln\frac{T_1}{T_0} = \nu R \ln\frac{V_1}{V_0} \Rightarrow C = C_V + \frac{\nu R}{\ln\frac{T_1}{T_0}}\ln\frac{V_1}{V_0}$

$C_V = \frac{3}{2}R\,\frac{\nu}{1\,моль}$ $\quad pV = \nu RT \Rightarrow T_1 = \frac{T_0 p_1 V_1}{p_0 V_0};$

$\nu = \frac{p_0 V_0}{RT_0} \Rightarrow C = \frac{3}{2}R\frac{\nu}{1\,моль} + \frac{p_0 V_0 R}{RT_0 \ln\frac{p_1V_1}{p_0V_0}}\ln\frac{V_1}{V_0} \Rightarrow$ ~~Дж/К~~

$\Rightarrow C \approx 0{,}68\ \frac{Дж}{К}.$

№1.54.

ν = 1 моль; $p_0 S^2 = kV_0$. C - ? $dpS = kdx \Rightarrow$

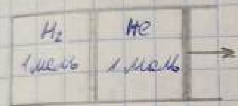
$\Rightarrow dpS = \frac{kdV}{S} \Rightarrow pS^2 = kV \Rightarrow \frac{dp}{p} = \frac{dV}{V}$.

$\frac{dp}{p} + \frac{dV}{V} = \frac{dT}{T} \Rightarrow \frac{dT}{T} = \frac{2dV}{V}$.

$$C = C_V + p\frac{dV}{dT} = C_V + \frac{pV}{2T} = C_V + \frac{R}{2}.$$

Ответ: $C = C_V + \frac{R}{2}$.

№1.87.



$V_1 = 2V_0$. T_1 - ?

p_0, $T_0 = 293$ K

$\delta Q = dU^{H_2} + dU^{He} + \delta A = 0 \Rightarrow$

$\Rightarrow C_V^{H_2} dT + C_V^{He} dT + p^{He} dV^{He} = 0$. $p = \frac{RT}{V} \Rightarrow$

$\Rightarrow (C_V^{H_2} + C_V^{He})dT = -RT\frac{dV^{He}}{V^{He}} \Rightarrow \left(\frac{C_V^{H_2} + C_V^{He}}{R}\right)\ln\frac{T_1}{T_0} = \ln\frac{V_1^{He}}{V_0^{He}}$

$C_V^{He} = \frac{3}{2}R$; $C_V^{H_2} = \frac{5}{2}R \Rightarrow 4\ln\frac{T_1}{T_0} = \ln 2 \Rightarrow T_1 = \frac{T_0}{\sqrt[4]{2}} \approx 246$ K.

Ответ: $T_1 = 246$ K.

№2.6.

$\mu_a = 40$ г/моль; $\mu_г = 4$ г/моль; $\nu_a = 0{,}01\nu_г$. $\frac{\Delta v_{зв}}{v_{зв}}$ - ?

$v_{зв\,г}^2 - v_{зв\,см}^2 = (v_{зв\,г} - v_{зв\,см})(v_{зв\,г} + v_{зв\,см}) \approx 2\Delta v_{зв} v_{зв\,г} \Rightarrow$

$\Rightarrow \frac{\Delta v_{зв}}{v_{зв}} \approx \frac{1}{2}\left(1 - \frac{v_{зв\,см}^2}{v_{зв\,г}^2}\right)$

$v_{зв\,см}^2 = \frac{c_{p\,см} RT}{\mu_{см} c_{V\,см}} = RT\left(\frac{m_г}{\mu_г(m_г + m_a)} + \frac{m_a}{\mu_a(m_г + m_a)}\right)\frac{\nu_г c_{pг} + \nu_a c_{pa}}{\nu_г c_{Vг} + \nu_a c_{Va}} =$

$= RT\frac{\nu_г + \nu_a}{\nu_г\mu_г + \nu_a\mu_a}\cdot\frac{\nu_г c_{pг} + \nu_a c_{pa}}{\nu_г c_{Vг} + \nu_a c_{Va}} \Rightarrow \frac{\Delta v_{зв}}{v_{зв}} \approx \frac{1}{2}\left(1 - \frac{\nu(1+0{,}01)\frac{5}{2}R(1+0{,}01)\mu_г}{\nu(\mu_г + 0{,}01\mu_a)\frac{3}{2}R(1+0{,}01)\frac{5}{3}}\right) =$

$= \frac{1}{2}\left(1 - \frac{1{,}01\mu_г}{\mu_г + 0{,}01\mu_a}\right) \Rightarrow \frac{\Delta v_{зв}}{v_{зв}} \approx \frac{1}{2}\left(1 - \frac{1{,}01}{1{,}1}\right) \approx 0{,}045$

Ответ: $\frac{\Delta v_{зв}}{v_{зв}} \approx 4{,}5\%$.

№1.100.

Дано: $\delta Q = -2dU$; p_1; $V_2 = \frac{V_1}{2}$. p_2 - ?

$\delta Q = CdT$; $dU = \frac{5}{2}RdT \Rightarrow$

$\Rightarrow CdT = -2\cdot\frac{5}{2}RdT \Rightarrow$

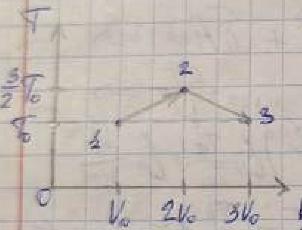
$\Rightarrow C = -5R = const \Rightarrow$ процесс политропический $\Rightarrow$

$\Rightarrow pV^n = const \Rightarrow p_2 = p_1\left(\frac{V_1}{V_2}\right)^n = p_1\cdot 2^n$.

$n = \frac{-5R - \frac{7}{2}R}{-5R - \frac{5}{2}R} = \frac{17}{15} \Rightarrow p_2 = 2^{\frac{17}{15}} p_1$

Ответ: $p_2 \approx 2{,}19\, p_1$.

T1.



ΔC - ? $C = \frac{\delta Q}{dT} = C_V + \frac{p\,dV}{dT} =$

$= C_V + \frac{RT\,dV}{V\,dT}$

1-2: $\frac{dV}{dT} = 2\frac{V_0}{T_0}$; 2-3: $\frac{dV}{dT} = -2\frac{V_0}{T_0}$.

$C_{2-} = C_V + 2\frac{V_0}{T_0}\cdot\frac{\frac{3}{2}RT_0}{2V_0} = C_V + \frac{3}{2}R$;

$C_{2+} = C_2 - 2\frac{V_0}{T_0}\cdot\frac{\frac{3}{2}RT_0}{2V_0} = C_V - \frac{3}{2}R$.

$\Delta C = C_{2+} - C_{2-} = C_V - \frac{3}{2}R - C_V - \frac{3}{2}R = -3R$.

Ответ: $\Delta C = -3R$.

№1.75.

Дано: $\frac{m(He)}{m(H_2)} = \frac{2}{1}$; $p_1 = 8$ атм; $p_2 = 1$ атм; $T_1 = 600$ K. T_2 - ?

Процесс адиабатический $\Rightarrow$ ~~...~~ $p^{1-\gamma}T^{\gamma} = const$

$\gamma = \frac{C_p}{C_V} = \frac{\nu_в C_{pв} + \nu_г C_{pг}}{\nu_в C_{Vв} + \nu_г C_{Vг}} = \frac{\nu_в(C_{Vв}+R) + \nu_г(C_{Vг}+R)}{\nu_в C_{Vв} + \nu_г C_{Vг}}$

$\nu_в = \frac{1}{3}\frac{m}{\mu_в}$; $\nu_г = \frac{2}{3}\frac{m}{\mu_г}$. $\mu_г = 2\mu_в \Rightarrow \nu_в = \nu_г$.

$C_{V_1} = \frac{5}{2}R;\ C_{V_2} = \frac{3}{2}R \Rightarrow \gamma = \frac{5/2 + 2 + 3/2}{5/2 + 3/2} = \frac{3}{2} \Rightarrow$

$T_2 = T_1 \sqrt[\frac{3}{2}]{\frac{p_2^{-0,5}}{p_1^{-0,5}}} \Rightarrow T_2 = 300\,K.$

Ответ: $T_2 = 300\,K$.

№ 1.83

$t = 610°C;\ c_p = 0{,}14\ \frac{Дж}{г \cdot K};$
$A = 26\,г \quad \alpha - ?$

$c_p = \frac{\sum_i \nu_i C_{p_i}}{\sum_i \nu_i} \Rightarrow c_p = \frac{\frac{7}{2}(1-\alpha)R + \frac{5}{2}2\alpha R}{2(1-\alpha)A + 2\alpha A} \Rightarrow$

$\Rightarrow \alpha = \frac{4c_pA - 7R}{3R} \Rightarrow \alpha \approx 0{,}5.$

Ответ: $\alpha = 0{,}5$.

№ 4

$\frac{V_{max}}{V_{min}} = 2;\ i = 5.\ \eta - ? \quad 2: T_0;\ p_0,\ V_0 \quad 3: p_0,\ 2V_0 \Rightarrow 2T_0.$

$3: V_0,\ p_3,\ T_3. \quad Q_{31} = const \Rightarrow p_0(2V_0)^\gamma = p_3 V_0^\gamma \Rightarrow$

$\Rightarrow p_3 = 2^\gamma p_0. \quad \gamma = \frac{7}{5} \Rightarrow T_3 = 2^{\frac{7}{5}} T_0.$

$Q_{12} = C_p \nu R(T_0 - 2T_0) = \frac{-7}{2}\nu R T_0;$

$Q_{23} = C_V \nu R(2^{\frac{7}{5}}T_0 - T_0) = \frac{5}{2}\nu R T_0 (2^{\frac{7}{5}} - 1) \Rightarrow$

$\Rightarrow \eta = \frac{5(2^{\frac{7}{5}} - 1) - 7}{5(2^{\frac{7}{5}} - 1)} \approx 0{,}146.$

№ 5

$T_2 = \frac{3}{2}T_1;\ T_3 = \frac{3}{4}T_1;\ T_4 = \frac{1}{20}T_1.\ \eta - ? \quad Q = \int T\,dS \Rightarrow$

$\Rightarrow \eta$ можно найти через площадь под графиком.

$\eta = \frac{T_1 + T_2 - T_3 - T_4}{T_1 + T_2} = 1 - \frac{T_3 + T_4}{T_1 + T_2}.\ \eta = 1 - \frac{0{,}75 + \frac{1}{20}}{1 + 1{,}5} = 0{,}68.$

№ 6

$Q^- = 65\,Дж;\ Q^+ = 80\,Дж;\ T^+ = 320\,K.\ T^- - ? \quad \eta = \frac{Q^+ - Q^-}{Q^-} = \frac{T^+ - T^-}{T^-} \Rightarrow$

$\Rightarrow T^- = T^+ \frac{Q^-}{Q^+} \Rightarrow T^- = 320 \cdot \frac{65}{80} = 260\,K.$

№3.25.

$m_1 = 1$ кг; $m_2 = 1$ кг; $T_1 = 373$ K; $T_2 = 273$ K; $q = 80 \frac{ккал}{кг}$. A - ? T_k - ?

Для получения максимальной работы будем считать процесс идеальным $\Rightarrow \frac{|\delta Q_н|}{T_н} = \frac{\delta Q_х}{T_2}$; $\delta Q_х = q\,dm_н$;

$\delta Q_н = c m_1 dT_н \Rightarrow \frac{c m_1 dT_н}{T_н} = \frac{q\,dm_н}{T_2} \Rightarrow dm_н = \frac{c m_1 T_2 dT_н}{q T_н} \Rightarrow$

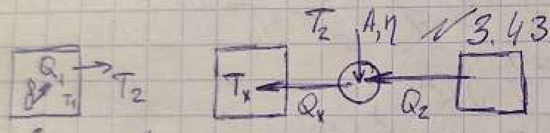
$\Rightarrow m_2 = \frac{c m_1 T_2}{q} \ln\frac{T_1}{T_k} \Rightarrow T_k = T_1 \exp\left(\frac{-q m_2}{c m_1 T_2}\right) \Rightarrow$

$\Rightarrow T_k \approx 278{,}3$ K. $A = Q_н - |Q_х| = c m_1 (T_1 - T_k) -$

$- q m_2 \Rightarrow A \approx 62$ кДж.

Ответ: $T_k = 278$ K; $A = 62$ кДж.

№3.43.



$T_1 = 270$ K, $T_2 = 250$ K, $\eta = \frac{A}{Q_1} = 0{,}4$. T_x - ?

$\dot{Q}_1 = \alpha(T_1 - T_2)$; $Q_x = Q_2 + A = Q_2 + \eta Q_1$.

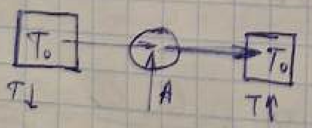
$\frac{Q_2}{T_2} = \frac{Q_x}{T_x} \Rightarrow Q_2 = Q_x \frac{T_2}{T_x}$. $\dot{Q}_x = \dot{Q}_2 + \eta \dot{Q}_1 \Rightarrow$

$\Rightarrow \alpha(T_x - T_2) = \alpha \frac{T_2}{T_x}(T_x - T_2) + \alpha\eta(T_1 - T_2) \Rightarrow$

$\Rightarrow T_x^2 - T_x\left(2T_2 + \eta(T_1 - T_2)\right) + T_2^2 = 0 \Rightarrow \left[\begin{array}{l} T_x = 299 \text{ K} \\ T_x = 209 \text{ K} \end{array}\right.$

Ответ: $T_x = 299$ K или 209 K.

Т2.



$T_0 = 300$ K; $T_1 = 200$ K; $\nu = 1$ моль. T_2 - ? A - ?

$\delta Q = dU + p\,dV$; $pV = RT$.

$T dS = C_V dT + \frac{RT}{V} dV \Rightarrow \Delta S = C_V \ln\frac{T_1}{T_0} + R \ln\frac{V_1}{V_0} \Rightarrow$

$\Rightarrow \Delta S = C_v \ln\frac{T_1}{T_0} + C_v \ln\frac{T_2}{T_0} = 0 \Rightarrow \ln\frac{T_1}{T_0} = -\ln\frac{T_2}{T_0} \Rightarrow T_0^2 = T_1 T_2 \Rightarrow$

$\Rightarrow T_2 = \frac{T_0^2}{T_1} = 450$ K.

$A = |Q_2| - |Q_1| = C_v(T_2 - T_0) - C_v(T_0 - T_1) = C_v(T_2 + T_1 - 2T_0) \Rightarrow$

$\Rightarrow A = \frac{5}{2}R(450 + 200 - 600) \approx 1037$ Дж.

Ответ: $T_2 = 450$ K; $A = 1037$ Дж.

№ 4.80.

$\frac{\rho}{\rho_0} = 2$; $H = 12{,}2$ км; $g = 8{,}87\ \frac{м}{с^2}$; $C_v = 5R$. T_0 – ?

Атмосфера адиабатическая $\Rightarrow pV^\gamma = const \Rightarrow p\rho^{-\gamma} = const$.

$p = \rho g h \Rightarrow dp = -\rho g dz$; $p = \alpha \rho^\gamma \Rightarrow$

$\Rightarrow dp = \alpha\gamma\rho^{\gamma-1} d\rho = -\rho g dz \Rightarrow \alpha\gamma\rho^{\gamma-2} d\rho = -g dz$.

$\alpha\frac{\gamma}{\gamma-1}\rho^{\gamma-1} = -gz + const$. При $z = 0$ $\rho = \rho_0 \Rightarrow$

$\Rightarrow \alpha\frac{\gamma}{\gamma-1}\rho^{\gamma-1} = -gz + \alpha\frac{\gamma}{\gamma-1}\rho_0^{\gamma-1} \Rightarrow$

$\Rightarrow \alpha\frac{\gamma}{\gamma-1}\rho_0^{\gamma-1}\left(\left(\frac{\rho}{\rho_0}\right)^{\gamma-1} - 1\right) = -gz$.

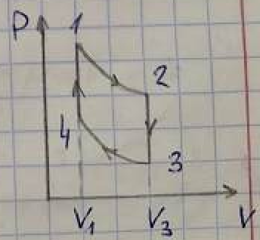
$\frac{p}{\rho} = \frac{RT}{\mu} \Rightarrow \alpha\rho_0^{\gamma-1} = \frac{RT_0}{\mu} \Rightarrow \alpha = \frac{RT_0}{\mu\rho_0^{\gamma-1}} \Rightarrow$

$\Rightarrow \frac{\gamma}{\gamma-1}\frac{RT_0}{\mu}\left(\left(\frac{\rho}{\rho_0}\right)^{\gamma-1} - 1\right) = -gH \Rightarrow T_0 = \frac{\mu g H}{R\left(1 - \left(\frac{\rho}{\rho_0}\right)^{\gamma-1}\right)} \cdot \frac{\gamma-1}{\gamma}$.

$C_p = C_v + R$; $\gamma = \frac{C_p}{C_v} \Rightarrow T_0 = \frac{\mu g H}{6R\left(1 - \left(\frac{\rho}{\rho_0}\right)^{\gamma-1}\right)} \Rightarrow T_0 \approx 740$ K.

Ответ: $T_0 \approx 740$ K.

№ 3.52.



$T_{max} = T_1$; $T_{min} = T_3$. A – ? $A = Q_н - |Q_х|$

$Q_н = C_v(T_1 - T_4)$; $Q_х = C_v(T_3 - T_2) \Rightarrow$

$\Rightarrow A = C_V(T_1 - T_4 + T_3 - T_2)$; $C_V = \frac{3}{2}R$. $\frac{V_1}{V_3} = \alpha$

$T_1 V_1^{\gamma-1} = T_2 V_3^{\gamma-1} \Rightarrow T_2 = T_1\left(\frac{V_1}{V_3}\right)^{\gamma-1} = T_1\alpha^{\gamma-1}$;

$T_3 V_3^{\gamma-1} = T_4 V_1^{\gamma-1} \Rightarrow T_4 = T_3\left(\frac{V_3}{V_1}\right)^{\gamma-1} = T_3\alpha^{1-\gamma} \Rightarrow$

$\Rightarrow A = C_V\left(T_1\left(1-\alpha^{\gamma-1}\right) + T_3\left(1-\alpha^{1-\gamma}\right)\right) \Rightarrow$

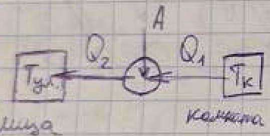
$\Rightarrow -T_1(\gamma-1)\alpha^{\gamma-2} - T_3(1-\gamma)\alpha^{-\gamma} = 0 \Rightarrow T_1\alpha^{\gamma-2} = T_3\alpha^{-\gamma} \Rightarrow$

$\Rightarrow \frac{T_1}{T_3} = \alpha^{2-2\gamma} \Rightarrow \alpha = \left(\frac{T_1}{T_3}\right)^{\frac{1}{2-2\gamma}} \Rightarrow$ т.к. $\gamma = \frac{5}{3}$, $\alpha = \left(\frac{T_1}{T_3}\right)^{-\frac{3}{4}} \Rightarrow$

$\Rightarrow A = C_V\left(T_1\left(1-\left(\frac{T_1}{T_3}\right)^{-\frac{2\cdot3}{3\cdot4}}\right) + T_3\left(1-\left(\frac{T_1}{T_3}\right)^{\frac{2\cdot3}{3\cdot4}}\right)\right) = C_V\left(T_1 - \sqrt{T_3T_1} + T_3 - \sqrt{T_3T_1}\right) = \frac{3}{2}R\left(\sqrt{T_1}-\sqrt{T_3}\right)^2$.

Ответ: $A = \frac{3}{2}R\left(\sqrt{T_1}-\sqrt{T_3}\right)^2$.

№3.47.



$t_к = 21°C$; $t_1 = 26°C$; $\Delta t = 5°C$.

$t_2 = t_1 + \Delta t$. $\frac{A_2}{A_1}$ - ?

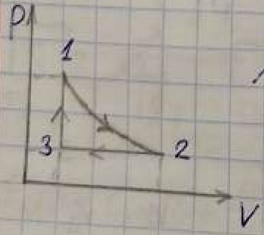
$Q_2 = A + Q_1$. $Q_1 = \alpha(T_{ул} - T_к)$; $\frac{Q_2}{T_{ул}} = \frac{Q_1}{T_к} \Rightarrow$

$\Rightarrow A = Q_2 - Q_1 = Q_1\left(\frac{T_{ул}}{T_к} - 1\right) = \frac{\alpha}{T_к}\left(T_{ул} - T_к\right)^2 \Rightarrow$

$\Rightarrow \frac{A_2}{A_1} = \left(\frac{T_2 - T_к}{T_1 - T_к}\right)^2 \Rightarrow \frac{A_2}{A_1} = 4$.

Ответ: $\frac{A_2}{A_1} = 4$.

№4.15.



$b = \frac{\Delta S_{max}}{C_V} = 0{,}2$; $\gamma = \frac{C_p}{C_V}$. η - ?

$\oint\frac{\delta Q}{T} = 0 \Rightarrow C_p\int_{T_2}^{T_3}\frac{dT}{T} + C_V\int_{T_3}^{T_1}\frac{dT}{T} = 0 \Rightarrow$

$\Rightarrow C_p\ln\frac{T_3}{T_2} + C_V\ln\frac{T_1}{T_3} = 0 \Rightarrow$

$$\frac{C_p}{C_v}=\gamma=-\frac{\ln\frac{T_1}{T_3}}{\ln\frac{T_3}{T_2}}\Rightarrow \gamma\ln\frac{T_3}{T_2}=\ln\frac{T_3}{T_1}\Rightarrow\left(\frac{T_3}{T_2}\right)^{\gamma}=\frac{T_3}{T_1}.$$

$$\Delta S_{max}=-\int_{T_2}^{T_3}C_p\frac{dT}{T}=C_v\int_{T_3}^{T_1}\frac{dT}{T}\Rightarrow\Delta S_{max}=C_v\ln\frac{T_1}{T_3}\Rightarrow$$

$$\Rightarrow\ln\frac{T_1}{T_3}=b=\gamma\ln\frac{T_2}{T_3}.\qquad \eta=1-\frac{|Q_x|}{Q_н}.$$

$$Q_x=C_p(T_3-T_2);\ Q_н=C_v(T_1-T_3)\Rightarrow\eta=1-\gamma\frac{T_2-T_3}{T_1-T_3}.$$

$$T_1=T_3e^{b};\ T_2=T_3e^{\frac{b}{\gamma}}\Rightarrow\eta=1-\gamma\frac{e^{\frac{b}{\gamma}}-1}{e^{b}-1}\Rightarrow\eta\approx0{,}0254.$$

Ответ: $\eta=0{,}0254$.

№4.73.

$C_1=\alpha T;\ C_2=\beta\sqrt{T};\ 2\beta=3\alpha\sqrt{T_1};\ T_1=T_{min}.\ \eta-?$

$$dS=C\frac{dT}{T}\Rightarrow\Delta S=\alpha\int_{T_1}^{T_2}dT=-\beta\int_{T_2}^{T_1}\frac{dT}{\sqrt{T}}\Rightarrow$$

$$\Rightarrow\alpha(T_2-T_1)=2\beta(\sqrt{T_2}-\sqrt{T_1})\Rightarrow T_2-T_1=3\sqrt{T_1T_2}-3T_1\Rightarrow$$

$$\Rightarrow T_2-3\sqrt{T_1T_2}+2T_1=0.\qquad D=9T_1-8T_1=T_1\Rightarrow\sqrt{T_2}=\frac{3\sqrt{T_1}\pm\sqrt{T_1}}{2}\Rightarrow$$

$$\Rightarrow T_2=4T_1\qquad\eta=1-\frac{|Q_2|}{Q_1}.\quad Q_1=\alpha\int_{T_1}^{T_2}T\,dT=$$

$$=\alpha\left(\frac{T_2^2-T_1^2}{2}\right)=\alpha\frac{15T_1^2}{2}.\qquad Q_2=\beta\int_{T_2}^{T_1}\sqrt{T}\,dT=$$

$$=\frac{2}{3}\beta\left(T_1^{\frac{3}{2}}-T_2^{\frac{3}{2}}\right)=\alpha\sqrt{T_1}\left(T_1^{\frac{3}{2}}-8T_1^{\frac{3}{2}}\right)=-7\alpha T_1^2\Rightarrow$$

$$\Rightarrow\eta=1-\frac{7\cdot2\alpha T_1^2}{15\alpha T_1^2}=\frac{1}{15}.$$

Ответ: $\eta=\frac{1}{15}$.

№7.

$m=10$ г; $V_2=2V_1$. $\Delta S-?$ $\qquad TdS=\nu C_v dT+p\,dV\Rightarrow$

$$\Rightarrow dS=\nu C_v\frac{dT}{T}+\frac{p}{T}dV=\nu C_v\frac{dT}{T}+\nu R\frac{dV}{V}.$$

$$T=const\Rightarrow\Delta S=\nu R\ln\frac{V_2}{V_1}\Rightarrow\Delta S=\frac{10}{2}\cdot8{,}31\ln2\approx28{,}8\ \frac{Дж}{К}.$$

№8.

$m_л = 90г$, $t_л = 0°C$, $m_к = 330г$, $t_к = 100°C$, $\lambda = 330 \frac{Дж}{г}$, $C = 0{,}9 \frac{Дж}{г \cdot К}$. ΔS-?

Тепло для таяния всего льда: $Q_л = \lambda m_л = 29\,700$ Дж;

Тепло для охлажд. кастрюли до 273К: $Q_к = C m_к (T_к - T_н) = -29\,700$ Дж;

$\Rightarrow$ В момент равновесия $T_к = 273К$, весь лёд растает.

$\Delta S = \Delta S_л + \Delta S_к$. $\Delta S_л = \frac{Q_л}{T_л}$; $\Delta S_к = \int \frac{c m_к \, dT}{T} = c m_к \ln \frac{T_к}{T_н} \Rightarrow$

$\Rightarrow \Delta S = 108{,}8 - 92{,}7 = 16{,}1 \frac{Дж}{К}$.

№9.

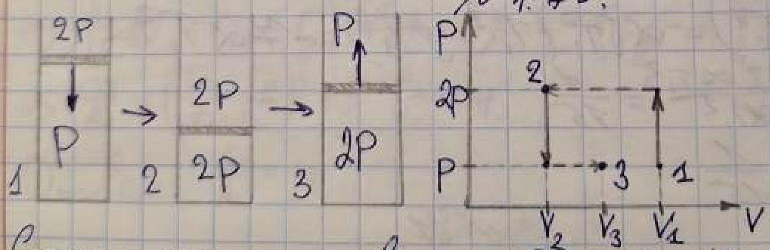
$m = 1$ кг, $T = 298 K = const$, $p_1 = 1$ мбар; $p_2 = 2$ мбар. $\Delta F, \Delta G$-?

$dF = dU - TdS$; $dG = dU + \cancel{\nu R dT}^{0} - TdS \Rightarrow$

$\Rightarrow dF = dG = dU - TdS = \cancel{\nu C_V dT}^{0} - TdS = -Q = -\cancel{dU}^{0} - pdV = -\nu RT \frac{dV}{V} \Rightarrow \Delta F = \Delta G = \nu RT \ln \frac{V_1}{V_2} =$

$= \nu RT \ln \frac{p_2}{p_1} \Rightarrow \Delta F = \Delta G = \frac{1}{0{,}018} \cdot 8{,}31 \cdot 298 \ln 2 \approx 95{,}4$ (кДж)

№4.75.



Сосуд теплоизолирован $\Rightarrow Q = TdS = dU + pdV = 0$.

Сжатие: $C_V (T_2 - T_1) = -2p (V_2 - V_1) \Leftrightarrow C_V T_1 \left(\frac{T_2}{T_1} - 1\right) = 2pV_1 \left(1 - \frac{V_2}{V_1}\right)$

$\frac{pV_1}{T_1} = \frac{2pV_2}{T_2} \Rightarrow \frac{T_2}{T_1} = \frac{2V_2}{V_1}$. $\frac{3}{2} RT_1 \left(\frac{2V_2}{V_1} - 1\right) = 2RT_1 \left(1 - \frac{V_2}{V_1}\right) \Rightarrow$

$\Rightarrow 3\frac{V_2}{V_1} - \frac{3}{2} = 2 - 2\frac{V_2}{V_1} \Rightarrow \frac{V_2}{V_1} = 0{,}7$; $\frac{T_2}{T_1} = 1{,}4$.

Расширение: $C_V(T_3 - T_2) = -p(V_3 - V_2) \Leftrightarrow C_V T_2\left(\frac{T_3}{T_2} - 1\right) = pV_2\left(1 - \frac{V_3}{V_2}\right)$

$\frac{2pV_2}{T_2} = \frac{pV_3}{T_3} \Rightarrow \frac{T_3}{T_2} = \frac{V_3}{2V_2} \Rightarrow \frac{5}{2}R \cdot \frac{2pV_2}{R}\left(\frac{V_3}{2V_2} - 1\right) = pV_2\left(1 - \frac{V_3}{V_2}\right) \Rightarrow$

$\Rightarrow \frac{5}{2}\frac{V_3}{V_2} - 5 = 1 - \frac{V_3}{V_2} \Rightarrow \frac{V_3}{V_2} = \frac{8}{5};\ \frac{T_3}{T_2} = \frac{4}{5}$

$\Delta S = C_V \ln\frac{T_2}{T_1} + R\ln\frac{V_2}{V_1} + C_V\ln\frac{T_3}{T_2} + R\ln\frac{V_3}{V_2} =$

$= C_V \ln\frac{7\cdot 4}{5\cdot 5} + R\ln\frac{7\cdot 8}{10\cdot 5} = C_p \ln\frac{28}{25} \approx 2{,}5\ \frac{Дж}{К}.$

№ 5.75.

$G(p,T) = aT(1 - \ln T) + RT\ln p - TS_0 + U_0;$

$a, R, S_0, U_0 = const;\quad U(V,T),\ H(V,T) - ?\quad a - ?$

$U = H - pV;\ H = G + ST \Rightarrow U = G + ST - pV.$

$G = pV - ST \Rightarrow V = \left(\frac{\partial G}{\partial p}\right)_T = \frac{RT}{p}.\quad S = -\left(\frac{\partial G}{\partial T}\right)_p = S_0 - R\ln p + a\ln T \Rightarrow$

$\Rightarrow \underline{H} = G + T(S_0 - R\ln p + a\ln T) = \underline{aT + U_0.}$

$\underline{U} = H - RT = aT + U_0 - RT = \underline{U_0 + T(a - R).}$

$dH = TdS + Vdp \Rightarrow$ при $p = const \hookrightarrow dH = TdS = \delta Q =$

$= C_p dT \Rightarrow \underline{C_p} = \left(\frac{\partial H}{\partial T}\right)_p \underline{= a.}$

№ 5.38.

$F = F_0 - \frac{\alpha B^2}{T};\ T = const;\ B_1 = B_0;\ B_2 = 0.\quad \Delta Q - ?$

$\Delta Q = T\Delta S;\quad \Delta S = -\left(\frac{\partial F}{\partial T}\right)_V = -\frac{\alpha B_0^2}{T^2} \Rightarrow \Delta Q = \frac{-\alpha B_0^2}{T}.$

№ 4.43.

Дано: $V_1^{O_2} = 3\,л;\ V_2^{N_2} = 2\,л;\ \nu_1 = \nu_2 = \nu = 0{,}5\,моль;\ T = 300\,K = const.\quad A_{max} - ?$

$TdS = \cancel{\nu C_V dT}^{\,0} + \delta A \Rightarrow A_{max} = T\Delta S.\quad \Delta S = \nu\left(\cancel{C_V \ln\frac{T_2}{T_1}}^{\,0} + R\ln\frac{V_к}{V_н}\right) \Rightarrow$

$\Rightarrow \cancel{\Delta S = 2\nu R \ln} \; \Delta S = \nu R \ln \frac{(V_1+V_2)^2}{V_1 V_2} \Rightarrow A_{max} = \nu R T \ln \frac{(V_1+V_2)^2}{V_1 V_2} \Rightarrow$

$\Rightarrow A_{max} \approx 1{,}8$ кДж.

№ 4.44.

Те же условия, но $T_1 = 300K \neq const$, $Q = const$. A_{max} - ?

$\Delta Q = \Delta U + \Delta A^{0} \Rightarrow A_{max} = -\Delta U_{max} = 2\nu C_v (T_1 - T_2)$

$p_{1н} V_1^{\gamma} = p_{1к} (V_1+V_2)^{\gamma} \Rightarrow p_{1к} = p_{10} \left(\frac{V_1}{V_2+V_1}\right)^{\gamma} = \frac{\nu R T_1}{V_1} \left(\frac{V_1}{V_1+V_2}\right)^{\gamma};$

$p_{2к} = \frac{\nu R T_1}{V_2} \left(\frac{V_2}{V_1+V_2}\right)^{\gamma}.$ $\quad p_к = p_{1к} + p_{2к} = \nu R T_1 \left(\frac{1}{V_1}\left(\frac{V_1}{V_1+V_2}\right)^{\gamma} + \frac{1}{V_2}\left(\frac{V_2}{V_1+V_2}\right)^{\gamma}\right)$

$p_к (V_1+V_2) = 2\nu R T_2 \Rightarrow T_2 = \frac{p_к (V_1+V_2)}{2\nu R} \Rightarrow$

$\Rightarrow A_{max} = 2\nu C_v \left(T_1 - \frac{T_1 (V_1+V_2)}{2} \left(\frac{1}{V_1}\left(\frac{V_1}{V_1+V_2}\right)^{\gamma} + \frac{1}{V_2}\left(\frac{V_2}{V_1+V_2}\right)^{\gamma}\right)\right), \quad \gamma = \frac{7}{5} \Rightarrow$

$\Rightarrow A_{max} \approx 1{,}55$ кДж.

№ 4.47.

Дано: $\nu = 10$ моль, $T_1 = 300K$, $T = 200K$, $A = 15$ кДж. ΔS-?

Т.к. до и после всех изменений газ находится в равновесном состоянии, ΔS при расширении и сжатии совпадают. $A = A_1 + A_2$;

$A_1 = T \Delta S; \quad A_2 = \Delta U = \nu C_v (T_1 - T) \Rightarrow$

$\Rightarrow A = T \Delta S + \nu C_v (T_1 - T) \Rightarrow \Delta S = \frac{A}{T} - \nu C_v \frac{T_1 - T}{T} \approx 12{,}65 \frac{Дж}{К}.$

Т 3.

Дано: N_2, $\nu = 1$ моль, $\frac{p_1}{p_2} = 3$. ΔS - ?

$\Delta S = C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1}; \quad C_v = \frac{5}{2} R.$

x - смещение пружины, S - площадь поршня.

ЗСЭ: $C_V T_1 = C_V T_2 + \frac{kx^2}{2}$. $\quad P_2 = \frac{kx}{S} \Rightarrow S = \frac{kx}{P_2}$;

$V_2 - V_1 = xS = \frac{kx^2}{P_2} \Rightarrow \frac{5}{2}RT_1 = \frac{5}{2}RT_2 + \frac{P_2}{2}(V_2 - V_1) \Rightarrow$

$\Rightarrow 5R(T_1 - T_2) = RT_2 - \frac{RT_1}{3} \Rightarrow \frac{16}{3}T_1 = 6T_2 \Rightarrow \frac{T_2}{T_1} = \frac{8}{9}$.

$\frac{3P_2V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow \frac{V_2}{V_1} = \frac{3T_2}{T_1} = \frac{8}{3} \Rightarrow$

$\Rightarrow \Delta S = \frac{5}{2}R\ln\frac{8}{9} + R\ln\frac{8}{3} \approx 0{,}69.$

№ 5.32.

$F = -\frac{RT}{2}\ln(AT^3V^2) \quad A = const. \quad C_p - ?$

$C_p = \left(\frac{\delta Q}{\partial T}\right)_p = \left(\frac{T\,\partial S}{\partial T}\right)_p = T\left(\frac{\partial S}{\partial T}\right)_p; \quad S = -\left(\frac{\partial F}{\partial T}\right)_V \Rightarrow$

$\Rightarrow S = \frac{R}{2}\ln(AT^3V^2) + \frac{RT \cdot 3AT^2V^2}{2AT^3V^2} = \frac{R}{2}\ln\left(\frac{AT^5R^2}{p^2}\right) + \frac{3}{2}R$

$C_p = \frac{p^2}{AT^5R^2} \cdot \frac{RT}{2} \cdot \frac{5AT^4R^2}{p^2} = \frac{5}{2}R$

Ответ: $C_p = \frac{5}{2}R$.

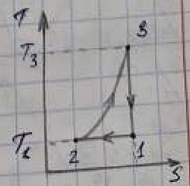
№ 5.54.

$f = aT\left(\frac{L}{L_0} - \left(\frac{L_0}{L}\right)^2\right)$; $a = 0{,}013\,\frac{Н}{К}$; $L_0 = 1$ м; $T_1 = 300$ К; $L = 2$ м; $C_L = 1{,}2\,\frac{Дж}{К}$.

$dF = -SdT - pdV. \quad pdV = \delta A = -fdL \Rightarrow$

$\Rightarrow dF = -SdT + fdL \Rightarrow S = \left(\frac{\partial F}{\partial T}\right)_L; \; f = \left(\frac{\partial F}{\partial L}\right)_T.$

Из соотношений Коши: $\left(\frac{\partial f}{\partial T}\right)_L = -\left(\frac{\partial S}{\partial L}\right)_T \Rightarrow dS = \frac{C_L dT}{T} - \left(\frac{\partial f}{\partial T}\right)_L dL.$



1-2: $T_1 = const$; $L_0 \to L \Rightarrow$

$\Rightarrow \Delta S_{12} = -\int_{L_0}^{L}\left(\frac{\partial f}{\partial T}\right)_L dL = -\int_{L_0}^{L} a\left(\frac{L}{L_0} - \left(\frac{L_0}{L}\right)^2\right)dL =$

$= -a\left(\frac{L^2 - L_0^2}{2L_0} + L_0^2\left(\frac{1}{L} - \frac{1}{L_0}\right)\right)$

$Q_{12} = T_1 \Delta S_{12} \Rightarrow Q_{12} \approx -3{,}9$ Дж.

2-3: $L = const$, $T_1 \to T_3 \Rightarrow \Delta S_{23} = \int_{T_1}^{T_3} \frac{C_L dT}{T} = C_L \ln\frac{T_3}{T_2}$;

$Q_{23} = C_L (T_3 - T_1)$

3-1: $Q = const \Rightarrow \Delta S = \Delta Q = 0 \Rightarrow \Delta S_{12} = -\Delta S_{23} \Rightarrow$

$\Rightarrow C_L \ln\frac{T_3}{T_1} = -\Delta S_{12} \Rightarrow T_3 = T_1 e^{\frac{\Delta S_{12}}{C_L}} \Rightarrow T_3 \approx 303{,}3\ K \Rightarrow$

$\Rightarrow Q_{23} = 3{,}96$ Дж.

№ 1.3.

Дано: $t = 0°C$; $\alpha = 1{,}8 \cdot 10^{-4}\ °C^{-1}$; $\beta = 3{,}9 \cdot 10^{-6}$ атм$^{-1}$. $p_0 = 1$ атм. $\lambda - ?$

$\left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V \left(\frac{\partial p}{\partial V}\right)_T = -1$; $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$; $\beta = \frac{-1}{V}\left(\frac{\partial V}{\partial p}\right)_T$; $\lambda = \frac{1}{p_0}\left(\frac{\partial p}{\partial T}\right)_V \Rightarrow$

$\Rightarrow \alpha V \cdot \frac{1}{\lambda p_0} \cdot \frac{-1}{\beta V} = -1 \Rightarrow \lambda = \frac{\alpha}{\beta p_0} \Rightarrow \lambda \approx 46\ °C^{-1}$.

№ 10.

$a = 1{,}3 \cdot 10^{-2}\ \frac{Н}{К}$; $l_0 = 1$ м; $l = 2$ м; $T = 300\ K = const$. $\Delta F - ?$

$f = aT\left(\frac{l}{l_0} - \left(\frac{l_0}{l}\right)^2\right)$. $dF = -S\cancel{dT}^{0} - \delta A$.

$\delta A = -f dl \Rightarrow \Delta F = \int_{l_0}^{l} f\, dl = aT\left(\frac{l^2}{2l_0} - \frac{l_0^2}{2l_0} + \frac{l_0^2}{l} - \frac{l_0^2}{l_0}\right) =$

$= aT\left(\frac{l^2}{2l_0} + \frac{l_0^2}{l} - \frac{3}{2}l_0\right) \Rightarrow \Delta F \approx 3{,}9$ Дж.

$\Delta U = 0$, т. к. $T = const$.

№ 11.

Дано: $m = 1$ г, $d = 2 \cdot 10^{-4}$ см; $\sigma = 26$ дин/см $= 2{,}6 \cdot 10^{-6}$ Н/м; $\rho = 0{,}9$ г/см^3; $T = const$.

$A - ?$ $A = E_2 - E_1 = \sigma(nS_2 - S_1)$. $S_2 = \pi d^2$;

$S_1 = 4\pi R_1^2$; $V_1 = \frac{4}{3}\pi R_1^3 = \frac{m}{\rho} \Rightarrow R_1 = \sqrt[3]{\frac{3}{4}\frac{m}{\pi\rho}} \Rightarrow S_1 = 4\pi\left(\frac{3m}{4\pi\rho}\right)^{2/3} =$

$= \left(\frac{3m}{\rho}\right)^{2/3}(4\pi)^{1/3} = \sqrt[3]{\frac{36\pi m^2}{\rho^2}}$. $n = \frac{V_1}{V_2} = \frac{m}{\rho} \cdot \frac{3 \cdot 2}{\pi d^3} \Rightarrow$

$$\Rightarrow A = \sigma\left(\frac{6m}{\pi\rho d^3}\pi d^2 - \sqrt[3]{\frac{36\pi m^2}{\rho^2}}\right) = \sigma\left(\frac{6m}{\rho d} - \sqrt[3]{\frac{36\pi m^2}{\rho^2}}\right) \Rightarrow A \approx 8{,}67\cdot 10^{-2}\ \text{Дж}.$$

№ 5.16.

Дано: $t_1 = 0^\circ C$; $p_2 = 100$ атм; $\alpha = 1{,}81\cdot 10^{-4}\ {}^\circ C^{-1}$;

$c_p = 0{,}033\ \frac{\text{кал}}{\text{г}\,{}^\circ C}$; $\rho = 13{,}6\ \text{г/см}^3$. $\Delta T - ?$

$$dS = \left(\frac{\partial S}{\partial T}\right)_p dT + \left(\frac{\partial S}{\partial p}\right)_T dp = 0 \qquad \left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p = -\alpha V.$$

$$\left(\frac{\partial S}{\partial T}\right)_p = \frac{1}{T}\left(\frac{T\,\partial S}{\partial T}\right)_p = \frac{1}{T}\left(\frac{\delta Q}{\partial T}\right)_p = \frac{C_p}{T} = \frac{c_p \rho V}{T};\quad \alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p \Rightarrow$$

$$\Rightarrow -\frac{c_p\rho V}{T}dT = -\alpha V dp \Rightarrow \left(\frac{\partial T}{\partial p}\right)_S = \frac{\alpha V T}{c_p \rho V} = \frac{\alpha T}{c_p\rho} \Rightarrow$$

$$\Rightarrow \frac{\Delta T}{p_2 - p_1} = \frac{\alpha T}{c_p \rho} \Rightarrow \Delta T = \frac{\alpha T}{c_p\rho}(p_2 - p_1) \Rightarrow \Delta T \approx -0{,}262\ K.$$

№ 15.28.

Дано: $p_1 = 1$ атм; $p_2 = 11$ атм; $\rho = 1{,}26\ \text{г/см}^3$; $\mu = 92\ \text{г/моль}$;

$\gamma = 1{,}1$. 1) $T = 293\ K = const$; $Q = -10$ Дж; 2) $\Delta Q = 0$, $A = 8{,}76$ мДж.

$\alpha - ?$ $\beta_T - ?$ $\left(\frac{\partial p}{\partial T}\right)_V - ?$

1. $$dS(T,p) = \left(\frac{\partial S}{\partial T}\right)_p dT + \left(\frac{\partial S}{\partial p}\right)_T dp, \qquad \delta Q = T dS;\ T = const \Rightarrow$$

$$\Rightarrow \delta Q = \left(\frac{\partial S}{\partial p}\right)_T T dp \Rightarrow \Delta Q = T\left(\frac{\partial S}{\partial p}\right)_T (p_2 - p_1) = T\left(\frac{\partial V}{\partial T}\right)_p (p_1 - p_2) \Rightarrow$$

$$\Rightarrow \alpha = \frac{\Delta Q}{T(p_1 - p_2)V} = \frac{\Delta Q\,\rho}{T(p_1-p_2)m} = \frac{\Delta Q\,\rho}{\mu\nu T(p_1 - p_2)} \Rightarrow \alpha \approx 4{,}7\cdot 10^{-4}\ \frac{1}{K};$$

2. $\Delta Q = 0$: $$\delta A = p\,dV = p\left(\frac{\partial V}{\partial p}\right)_S dp \Rightarrow A = \beta_S V (p_2^2 - p_1^2)/2 \Rightarrow$$

$$\Rightarrow \beta_S = \frac{2A}{V(p_2^2 - p_1^2)} \Rightarrow \beta_T = \gamma\beta_S = \frac{2A\rho\gamma}{\mu\nu(p_2^2 - p_1^2)} \Rightarrow \beta_T \approx 2{,}2\cdot 10^{-10}\ \frac{1}{\text{Па}}.$$

$$\left(\frac{\partial V}{\partial T}\right)_p\left(\frac{\partial T}{\partial p}\right)_V\left(\frac{\partial p}{\partial V}\right)_T = -1 \Rightarrow \alpha V\left(\frac{\partial T}{\partial p}\right)_V\frac{-1}{\beta_T V} = -1 \Rightarrow \left(\frac{\partial p}{\partial T}\right)_V = \frac{\alpha}{\beta_T} \approx 2{,}14\cdot 10^{6}\ \frac{\text{Па}}{K}.$$

Ответ: $\alpha = 4{,}7\cdot 10^{-4}\ \frac{1}{K}$; $\beta_T = 2{,}2\cdot 10^{-10}\ \frac{1}{\text{Па}}$;

$\left(\frac{\partial p}{\partial T}\right)_V = 2{,}14\cdot 10^{6}\ \frac{\text{Па}}{K}$.

№ 12.8.

$h=10^{-3}$ мм; $T=300$ K; $\frac{\partial\sigma}{\partial T}=\frac{0{,}15\ дин/см}{1\ К}=1{,}5\cdot10^{-3}\ \frac{Н}{м\cdot К}$

$\Delta T-?$ $\quad U=\left(\sigma-T\frac{\partial\sigma}{\partial T}\right)\Pi \Rightarrow dU=\left(\frac{\partial U}{\partial T}\right)_\Pi dT+\left(\frac{\partial U}{\partial \Pi}\right)d\Pi=$

$=C_\Pi dT+\sigma d\Pi-T\frac{\partial\sigma}{\partial T}d\Pi.$

$\delta Q=0=dU+\delta A=C_\Pi dT+\sigma d\Pi-T\frac{\partial\sigma}{\partial T}d\Pi-\sigma d\Pi\Rightarrow$

$\Rightarrow C_\Pi dT=T\frac{\partial\sigma}{\partial T}d\Pi;\quad C_\Pi=cm=c\rho h\Pi\Rightarrow$

$\Rightarrow \frac{dT}{T}=\frac{1}{C_\Pi}\left(\frac{\partial\sigma}{\partial T}\right)_\Pi d\Pi\Rightarrow\frac{dT}{T}=\frac{1}{c\rho h}\left(\frac{\partial\sigma}{\partial T}\right)\frac{d\Pi}{\Pi}\Rightarrow$

$\Rightarrow \ln\left(1+\frac{\Delta T}{T}\right)=\frac{1}{c\rho h}\left(\frac{\partial\sigma}{\partial T}\right)\ln\frac{\Pi_к}{\Pi_н}.$

$\Pi_к=2\Pi_н$; плёнка имеет 2 поверхности $\Rightarrow$

$\Rightarrow \ln\left(1-\frac{\Delta T}{T}\right)=\frac{2}{c\rho h}\left(\frac{\partial\sigma}{\partial T}\right)\ln 2\Rightarrow\Delta T\approx-0{,}015$ K.

№ 5.63.

$\frac{\Delta V}{V}=0{,}01$; $\frac{\Delta T}{T}=0{,}028$; $\alpha=5{,}7\cdot10^{-5}\ K^{-1}$; $c_V=0{,}23\ \frac{Дж}{г\cdot К}$; $\rho=10{,}5$ г/см³.

$\beta_T-?$ $\quad \delta Q=0\Rightarrow dS(T,V)=\left(\frac{\partial S}{\partial V}\right)_T dV+\left(\frac{\partial S}{\partial T}\right)_V dT=0\Rightarrow$

$\Rightarrow\frac{mc_V\,dT}{T}+\frac{\alpha\,dV}{\beta_T}=0\Rightarrow mc_V\int_T^{T+\Delta T}\frac{dT}{T}=\frac{-\alpha}{\beta_T}\int_V^{V-\Delta V}dV\Rightarrow$

$\Rightarrow mc_V\ln\left(1+\frac{\Delta T}{T}\right)=\frac{-\alpha}{\beta_T}(V-\Delta V-V)=\frac{\alpha\Delta V}{\beta_T}=\rho Vc_V\ln\left(1+\frac{\Delta T}{T}\right)\Rightarrow$

$\Rightarrow\beta_T=\frac{\alpha\Delta V}{\rho c_V\ln\left(1+\frac{\Delta T}{T}\right)V}\Rightarrow\beta_T\approx8{,}5\cdot10^{-12}$ Па$^{-1}$.

№ 12.9.

$r=5$ см; $\nu=0{,}1$ моль; $T=290$ K; $\sigma=70$ дин/см; $\frac{d\sigma}{dT}=-0{,}15\ \frac{дин}{см\cdot К}$. $\Delta T-?$

$F=U-TS\Rightarrow dF=dU-TdS-SdT=-SdT-\delta A=$

$=-SdT+\sigma d\Pi\Rightarrow\left(\frac{\partial F}{\partial T}\right)_\Pi=-S\Rightarrow F=U+T\left(\frac{\partial F}{\partial T}\right)_\Pi\Rightarrow$

$\Rightarrow$ т.к. при $T = const$ $F = \sigma \Pi \Rightarrow U = \Pi\left(\sigma - T\frac{d\sigma}{dT}\right)$.

$$\nu C_V \Delta T = U = \Pi\left(\sigma - T\frac{d\sigma}{dT}\right) = 2\cdot 4\pi r^2\left(\sigma - T\frac{d\sigma}{dT}\right) \Rightarrow$$

$$\Rightarrow \Delta T = \frac{8\pi r^2}{\nu C_V}\left(\sigma - T\frac{d\sigma}{dT}\right) \Rightarrow \Delta T \approx 3{,}14\cdot 10^{-3}\,K.$$

№12.38.

$r_0 = 5$ см, $\sigma = 30$ дин/см; $T = 300\,K = const$. $\Delta S - ?$

Поскольку пузыри не полностью одинаковы, система будет неустойчивой, и при уменьшении одного из них давление в нём будет возрастать, из-за чего он будет уменьшаться ещё сильнее, и в результате останется один пузырь.

$$\Delta p = \frac{4\sigma}{r} \Rightarrow \frac{4\sigma}{r_k}\cdot\frac{4}{3}\pi r_k^3 = \frac{4\sigma}{r_0}\cdot 2\cdot\frac{4}{3}\pi r_0^3 \Rightarrow r_k = \sqrt{2}\,r_0 \Rightarrow$$

$$\Rightarrow \Delta S = \nu R \ln\frac{V_k}{V_0} = \nu R \ln\frac{2^{3/2}}{1} = \frac{1}{2}\nu R \ln 2$$

$$\nu = \frac{2p_0V_0}{RT} = \frac{2\cdot 4\sigma\cdot 4r_0^3\pi}{3RTr_0} = \frac{32\pi\sigma r_0^2}{3RT} \Rightarrow \Delta S = \frac{16\pi\sigma r_0^2}{3T}\ln 2 \approx 2{,}8\cdot 10^{-6}\,Дж.$$

№5.42.

$F = -AVT^4$; $A = 2{,}52\cdot 10^{-15}\,\frac{г}{см\cdot с^2\cdot К^4}$; $p = 1$ атм; $V = 1$ л. $C_V - ?$

$$dF = -SdT - pdV \Rightarrow S = -\left(\frac{\partial F}{\partial T}\right)_V = 4AVT^3;$$

$$p = -\left(\frac{\partial F}{\partial V}\right)_T = AT^4 \Rightarrow T = \left(\frac{p}{A}\right)^{1/4} \qquad C_V = \left(\frac{\delta Q}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V =$$

$$= T\cdot 12AVT^2 = 12AV\cdot\left(\frac{p}{A}\right)^{3/4} = 12A^{1/4}Vp^{3/4} \Rightarrow C_V \approx 8{,}4\cdot 10^4\,\frac{эрг}{К}.$$

$$C_{V\,ид.г.} = \frac{3}{2}\nu R = \frac{3}{2}\frac{pV}{T} = \frac{3}{2}pV\left(\frac{p}{A}\right)^{-1/4} = \frac{3}{2}p^{3/4}A^{1/4}V \Rightarrow \frac{C_V}{C_{V\,ид.г.}} = 8.$$

Ответ: $C_V = 8{,}4\cdot 10^4\,\frac{эрг}{К}$; $C_V = 8\,C_{V\,ид.г.}$

Т 4.

$C_V = aVT^3$; K. $(C_p - C_V) - ?$ $C_V = \left(\frac{\delta Q}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V = aVT^3$

$\Rightarrow \left(\frac{\partial S}{\partial T}\right)_V = aVT^2 \Rightarrow S = \frac{aVT^3}{3} + const$; $K = -V\left(\frac{\partial p}{\partial V}\right)_T \Rightarrow$

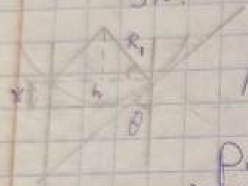
$\Rightarrow (\partial p)_T = -\frac{K}{V}(\partial V)_T$.

$C_p - C_V = T\left(\frac{\partial p}{\partial T}\right)_V\left(\frac{\partial V}{\partial T}\right)_p$; $\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T = \frac{aT^3}{3}$;

$\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T = \frac{V}{K}\cdot\frac{aT^3}{3} \Rightarrow C_p - C_V = T\cdot\frac{aT^3}{3}\cdot\frac{aT^3}{3}\cdot\frac{V}{K} =$

$= \frac{a^2T^7V}{9K}$.

№12.



$h = 0{,}1$ мм; $\theta = 60°$; $\sigma = 73\cdot10^{-3}$ Н/м. $x - ?$

$p_{пов} = \sigma\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$. $R_1 = \frac{h}{2\cos\theta}$; $R_2 \to +\infty$.

$\Rightarrow p_{пов} = \frac{2\sigma\cos\theta}{h} = \rho g x \Rightarrow x = \frac{2\sigma\cos\theta}{\rho g h} \Rightarrow x \approx 7{,}4$ см.

№13.

$t = 100\,°C$, $\Lambda = 40{,}7\ \frac{кДж}{моль}$. $u_п - u_ж - ?$

$Q = \Lambda\nu = \Delta U + p(V_п - V_в) \approx \Delta U + pV_п = \Delta U + \nu RT \Rightarrow$

$\Rightarrow u_п - u_ж = \frac{\Delta U}{\nu} = \Lambda - RT \Rightarrow u_п - u_ж \approx 37{,}6\ \frac{кДж}{моль}$.

№14.

$p_0 = 760$ мм рт. ст.; $T_0 = 373$ K; $p = 250$ мм рт. ст.; $\Lambda = 2{,}28\ \frac{кДж}{кг}$. $T - ?$

В равновесии $dG = Vdp - SdT = 0 \Rightarrow \frac{dp}{dT} = \frac{S}{V}$.

$V = \frac{\nu RT}{p}$; $Q = TS = \mu\Lambda\nu \Rightarrow \frac{dp}{dT} = \frac{\mu\Lambda\nu\cdot p}{T\cdot\nu RT} \Rightarrow$

$\Rightarrow \frac{dp}{p} = \frac{\mu\Lambda}{R}\,\frac{dT}{T^2} \Rightarrow \ln\frac{p}{p_0} = \frac{\mu\Lambda}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right) \Rightarrow T = \frac{1}{\frac{1}{T_0} - \frac{R}{\mu\Lambda}\ln\frac{p}{p_0}} \Rightarrow$

$\Rightarrow T \approx 344\ K = 71\ °C$.

№ 7.5.

$d = 1$ мм; $\sigma = 73 \cdot 10^{-3}$ Н/м; $t = 20\,°C$; $\cos\theta = 1$. $\frac{\Delta p}{p}$ – ?

$\Delta p = \rho_n g h$. $\rho_n = \frac{p\mu}{RT}$; Сечение капилляра круглое $\Rightarrow$

$\Rightarrow h = \frac{2\sigma\cos\theta}{\rho_в r g}$; $r = \frac{d}{2} \Rightarrow \frac{\Delta p}{p} = \frac{4 p \mu S}{p R T} \cdot \frac{\sigma \cos\theta}{\rho_в g d} = \frac{4\mu\sigma\cos\theta}{RT\rho_в d} \approx 2 \cdot 10^{-3}$.

№ 11.29.

$t = 100\,°C$; $\Delta T = 1$ K. $\frac{\Delta m}{m}$ – ?

$\Lambda \Delta m = C_n (m - \Delta m) \Delta T \approx m C_n \Delta T$.

$C_n = \frac{\delta Q}{dT} = \frac{dU}{dT} + p\frac{dV}{dT} = C_V + p\frac{dV}{dT}$. $pV = RT \Rightarrow$

$\Rightarrow p\,dV = R\,dT - V\,dp \Rightarrow C_n = C_V + R - V\frac{dp}{dT}$.

В равновесии $dG = V\,dp - S\,dT = 0 \Rightarrow \frac{dp}{dT} = \frac{S}{V} = \frac{\Lambda}{TV} \Rightarrow$

$\Rightarrow C_n = C_p - \frac{\Lambda}{T} \Rightarrow \frac{\Delta m}{m} = \frac{C_n \Delta T}{\Lambda} = \frac{\Delta T}{\Lambda}\left(C_p - \frac{\Lambda}{T}\right) = \frac{\Delta T}{T}\left(\frac{C_p T}{\Lambda} - 1\right) \approx 0{,}19\,\%$.

№ 11.16.

Дано: $V_0 = 5$ л; $m_в = 1$ кг; $t = 100\,°C$; $\Delta T = 1$ K; $\Lambda = 539$ кал/г

Δm – ? В равновесии $\frac{dp}{dT} = \frac{\Lambda m}{TV_n}$.

$p\,\cancel{dV_n}^{\,0} + V_n\,dp = \frac{R}{\mu}(m\,dT + T\,dm) =$

$= \frac{mRT}{\mu p}dp \Rightarrow \frac{dp}{p} = \frac{dT}{T} + \frac{dm}{m} \Rightarrow \frac{dp}{dT} = \frac{p}{T} + \frac{p\,dm}{m\,dT} = \frac{\Lambda m}{TV_n} \Rightarrow$

$\Rightarrow \frac{dT}{T}\left(\frac{\Lambda m}{V_n} - p\right) = \frac{p\,dm}{m} \Rightarrow \ln\left(1 + \frac{\Delta T}{T}\right)\left(\frac{\Lambda m}{pV_n} - 1\right) = \ln\left(1 + \frac{\Delta m}{m}\right) \rightsquigarrow$

$\rightsquigarrow \Delta m = \frac{m\Delta T}{T}\left(\frac{\Lambda m}{pV_n} - 1\right) = \frac{pV_n \Delta T \mu}{RT^2}\left(\frac{\Lambda\mu}{RT} - 1\right)$.

$p = 10^5$ Па; $V_n = V_0 - V_в = 4$ л $\Rightarrow \Delta m \approx 7{,}5 \cdot 10^{-5}$ кг.

№ 11.34.

Дано: $t = 0\,°C$; $\Delta Q = 0$; $p_1 = 1$ атм; $p_2 = 100$ атм; $v_в = 1\,\frac{см^3}{г}$; $v_л = 1{,}09\,\frac{см^3}{г}$; $C_л \approx 0{,}6\,C_в$

ΔT – ? $\frac{\Delta m}{m}$ – ?

$\Delta Q = \lambda \Delta m + C_л \Delta T (m - \Delta m) = 0 \Rightarrow$

$\Rightarrow c_л \Delta T m \approx -\lambda \Delta m \Rightarrow \frac{\Delta m}{m} = \frac{-c_л \Delta T}{\lambda}$. В равновесии

$\Delta G = 0 \Rightarrow \frac{\Delta p}{\Delta T} = \frac{\Delta S}{\Delta V}$, $\Delta V = \Delta m(v_л - v_в)$;

$\Delta S = \frac{-\lambda \Delta m}{T} \Rightarrow \frac{\Delta p}{\Delta T} = \frac{-\lambda \Delta m}{T \Delta m (v_л - v_в)} \Rightarrow \Delta T = \frac{-\Delta p T (v_л - v_в)}{\lambda} \Rightarrow$

$\Rightarrow \Delta T \approx -0{,}72\ K$; $\frac{\Delta m}{m} \approx \frac{-0{,}6 c_в \Delta T}{\lambda} \approx 0{,}0053$.

№12.51.

Дано: $\lambda = 2{,}26 \cdot 10^6 \frac{Дж}{кг}$; $t = 101^\circ C$; $p_0 = 1$ атм; $\sigma = 58{,}8 \frac{дин}{см^2}$; $v_п = 1{,}7 \frac{м^3}{кг}$; $t_0 = 100^\circ C$

$r - ?$ $\Delta p = \sigma\left(\frac{1}{r_1} + \frac{1}{r_2}\right) = \frac{2\sigma}{r}$

$\Delta G = 0 \Rightarrow \frac{\Delta p}{\Delta T} = \frac{S}{V} = \frac{\Lambda \mu R}{p V^2} =$

$= \frac{\Lambda p_0}{RT^2} = \frac{2\sigma}{r(t - t_0)} \Rightarrow r = \frac{2\sigma R T^2}{\Lambda p (t - t_0)} = \frac{2\sigma v T}{\Lambda (t - t_0)} \Rightarrow r \approx 0{,}08$ мм.

№11.74.

Дано: $S_ж = \frac{RT}{\theta}$; $\theta \approx 0{,}46\ K$; $S_{тв} = 0{,}7R$; $T_1 = 0{,}25\ K$; $T_2 = 0{,}1\ K$

$dG = 0 \Rightarrow \frac{dp}{dT} = \frac{\Delta S}{\Delta V}$. $\Delta S = S_ж - S_{тв} =$

$= R\left(\frac{T}{\theta} - 0{,}7\right) \Rightarrow dp = R\,dT \frac{1}{\Delta V}\left(\frac{T}{\theta} - 0{,}7\right)$

$\Rightarrow p_2 - p_1 = \frac{R}{\Delta V}\left(\frac{T_2^2 - T_1^2}{2\theta} - 0{,}7(T_2 - T_1)\right) \Rightarrow$

$\Rightarrow p_2 = \frac{R(T_2 - T_1)}{\Delta V}\left(\frac{T_2 + T_1}{2\theta} - 0{,}7\right) + p_1 \Rightarrow p_2 \approx 32{,}2$ атм.

№11.78.

Дано: $V = 10\ л$; $t_к = 0^\circ C$; $m = 6{,}7\ г$; $\lambda = 40{,}7 \frac{кДж}{моль}$

$\Delta T - ?$ $\frac{dp}{dT} = \frac{\Delta S}{\Delta V} = \frac{\lambda p}{RT^2} \Rightarrow \frac{dp}{p} = \frac{\lambda\, dT}{RT^2}$

$p\,dV + V\,dp = \nu R\,dT \Rightarrow \frac{dV}{V} + \frac{dp}{p} = \frac{dT}{T} \Rightarrow$

$\Rightarrow \frac{dV}{V} = \frac{dT}{T}\left(1 - \frac{\lambda}{RT}\right) \Rightarrow \frac{\Delta V}{V_0} = \frac{\Delta T}{T_0}\left(1 - \frac{\lambda}{RT_0}\right)$

$\Delta V = V - V_0 = V - \frac{mRT_0}{\mu p_0} \Rightarrow \Delta T = \left(V - \frac{mRT_0}{\mu p_0}\right)\frac{T_0}{V_0} \cdot \frac{1}{1 - \frac{\lambda}{RT_0}}$.

$\Delta T \approx 3{,}7\ K$.

Т5.

Дано: $\frac{m_л}{m_в}=0,1$; $q=330\ \frac{Дж}{кг}$; $\rho_в=1,00\ г/см^3$; $\rho_л=0,92\ г/см^3$; $\beta_в=4,8\cdot10^{-5}\ атм^{-1}$

$t-?$ $\quad \beta_в=\frac{-\Delta V}{\Delta p V}$ $\quad \Delta V=0,1V_0-V_л=$

$=0,1V_0-\frac{0,1V_0\rho_в}{\rho_л}=\frac{0,1V_0}{\rho_л}(\rho_л-\rho_в)$;

$V=0,9V_0 \Rightarrow \beta_в=\frac{0,1V_0}{\rho_л 0,9V_0\Delta p}(\rho_в-\rho_л)=\frac{1}{\Delta p 9\rho_л}(\rho_в-\rho_л)$.

$\frac{\Delta p}{\Delta T}=\frac{\Delta S}{\Delta \upsilon}=\frac{q}{T_0(\upsilon_в-\upsilon_л)}=\frac{q\rho_в\rho_л}{T_0(\rho_л-\rho_в)}$ (т.к. $\upsilon=\frac{1}{\rho}$) $\Rightarrow$

$\Rightarrow \Delta T=\frac{-T_0(\rho_в-\rho_л)^2}{9q\rho_в\rho_л^2\beta_в} \Rightarrow \Delta T\approx-1,45\ K \Rightarrow t\approx-1,45^\circ C$

№ 12.48.

Дано: $r=10^{-6}$ см; $T=293$ К; $\rho=13,56\ г/см^3$; $\sigma=487\ \frac{дин}{см}$; $\mu=200,6$ г/моль. $\frac{p_r}{p_\infty}-?$

Ф-ла Больцмана:

$p_r=p_\infty\cdot\exp\left(-\frac{\mu g h}{RT}\right)$

Из №12 $\Rightarrow h=\frac{2\sigma\cos\theta}{\rho r g} \Rightarrow \frac{p_r}{p_\infty}=\exp\left(-\frac{2\mu\sigma\cos\theta}{\rho r R T}\right)$.

$\cos\theta=-1 \Rightarrow \frac{p_r}{p_\infty}\approx 1,8$.

№ 16.

$\varphi=5$; $\tau=5,7$. $\pi-?$

$\left(\pi+\frac{3}{\varphi^2}\right)\left(\varphi-\frac{1}{3}\right)=\frac{8}{3}\tau \Rightarrow$

$\Rightarrow \left(\pi+\frac{3}{25}\right)\left(5-\frac{1}{3}\right)=\frac{8\cdot5,7}{3} \Rightarrow \frac{14\pi}{3}=15,2-\frac{14\cdot3}{3\cdot25} \Rightarrow \pi\approx3,137$.

№ 17.

$p_1=4$ атм; $p_2=1$ атм. $\Delta S-?$ $\quad H=const \Rightarrow dH=TdS+$

$+Vdp=0 \Rightarrow dS=-\frac{V}{T}dp=-R\frac{dp}{p} \Rightarrow \Delta S=R\ln\frac{p_1}{p_2} \Rightarrow$

$\Rightarrow \Delta S\approx 11,52\ \frac{Дж}{K}$.

№ 18.

(i – уд. энтальпия)

$\upsilon-?$ Из ур-ния Бернулли $\Rightarrow i+\frac{\upsilon^2}{2}=const. \Rightarrow$

$\Rightarrow i_1 = i_2 + \frac{v^2}{2} \Rightarrow v = \sqrt{2(i_1 - i_2)}$. $i = \frac{c_V T}{\mu} + \frac{RT}{\mu} = \frac{c_p T}{\mu} \Rightarrow$

$\Rightarrow v = \sqrt{\frac{2c_p}{\mu}(T_1 - T_2)}$ Условие адиабатически =

$\Rightarrow \frac{p_1^{\gamma-1}}{T_1^{\gamma}} = \frac{p_2^{\gamma-1}}{T_2^{\gamma}} \Rightarrow T_2 = T_1 \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow$

$\Rightarrow v = \sqrt{\frac{2}{\mu} c_p T_1 \left(1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right)}$. $p_2 \approx 0 \Rightarrow v = \sqrt{\frac{2}{\mu} c_p T_1}$.

$T_1 = 293\,K$ (н.у.); $c_p = \frac{7}{2}R \Rightarrow v = \sqrt{\frac{7}{\mu} R T_1} \Rightarrow v \approx 440$ м/с

T6.

Дано: $V_ж = V_п = 0{,}5 V_{кр}$; $\rho_ж = 1{,}9 \rho_{кр}$; $T_{кр} = 467\,K$. $T-?$

$m = const \Rightarrow \rho_{кр} = 0{,}5\rho_ж + 0{,}5\rho_п \Rightarrow \rho_п = 2\rho_{кр} - 1{,}9\rho_{кр} = 0{,}1\rho_{кр}$;

$\varphi_ж = \frac{V}{V_{кр}} = \frac{\rho_{кр}}{1{,}9\rho_{кр}} = \frac{1}{1{,}9}$; $\varphi_п = \frac{V}{V_{кр}} = \frac{\rho_{кр}}{0{,}1\rho_{кр}} = 10$

$\left(\pi + \frac{3}{\varphi^2}\right)\left(\varphi - \frac{1}{3}\right) = \frac{8}{3}\tau \Rightarrow$ $\begin{cases} \left(\pi + \frac{3}{\left(\frac{1}{1{,}9}\right)^2}\right)\left(\frac{1}{1{,}9} - \frac{1}{3}\right) = \frac{8}{3}\tau \\ \left(\pi + \frac{3}{10^2}\right)\left(10 - \frac{1}{3}\right) = \frac{8}{3}\tau \end{cases} \Rightarrow$

$\Rightarrow \tau \approx 0{,}8 = \frac{T}{T_{кр}} \Rightarrow T = 0{,}8 T_{кр} \Rightarrow T \approx 372\,K$.

№ 6.52.

Дано: $V_2 = 17 V_{кр}$, $C_V = 3R$; $\Delta Q = 0$. $\Delta S-?$

$dS(T,V) = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV = \frac{C_V}{T}dT + \left(\frac{\partial p}{\partial T}\right)_V dV$.

$\left(p + \frac{a}{V^2}\right)(V - b) = RT \Rightarrow p = \frac{RT}{V-b} - \frac{a}{V^2} \Rightarrow \left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{V-b}$. $\Rightarrow$

$\Rightarrow dS = \frac{C_V}{T}dT + \frac{R\,dV}{V-b} \Rightarrow \Delta S = C_V \ln\frac{T_2}{T_{кр}} + R\ln\frac{V_2 - b}{V_{кр} - b}$.

$V_2 = 17 V_{кр}$; $V_{кр} = 3b \Rightarrow \frac{V_2 - b}{V_{кр} - b} = \frac{50}{2} = 25$.

$dU(T,V) = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV = C_V dT + dV\left(\left(\frac{\partial p}{\partial T}\right)_V T - p\right) = C_V dT +$

$+ dV\left(\frac{RT}{V-b} - \frac{RT}{V-b} + \frac{a}{V^2}\right) = C_V dT + \frac{a\,dV}{V^2} \Rightarrow \Delta U = C_V(T_2 - T_{кр}) - a\left(\frac{1}{V_2} - \frac{1}{V_{кр}}\right)$

$= 0$, т.к. $\Delta Q = 0$ и $\Delta A = 0$ (расширение в вакуум). $\Rightarrow$

$\Rightarrow T_2 = \frac{a}{C_V}\left(\frac{1}{V_2}-\frac{1}{V_{кр}}\right)+T_{кр} \Rightarrow \frac{T_2}{T_{кр}} = 1-\frac{a}{T_{кр}3RV_{кр}}\left(1-\frac{1}{17}\right).$

$T_{кр}V_{кр} = 3b\cdot\frac{8a}{27bR}=\frac{8a}{9R} \Rightarrow \frac{T_2}{T_{кр}} = 1-\frac{3}{8}\cdot\frac{16}{17}=\frac{11}{17}\Rightarrow$

$\Rightarrow \Delta S = 3R\ln\frac{11}{17}+R\ln 25 \Rightarrow \Delta S \approx 15,9\ \frac{Дж}{К}.$

№ 2.11.

Дано: $T_1=273$ К; $v=400$ м/с; $p_2=1$ атм; $\Delta Q=0$. T_2, p_1 – ?

Из № 1.18: $v=\sqrt{\frac{2C_p}{\mu}(T_1-T_2)} \Rightarrow T_2=T_1-\frac{\mu v^2}{2C_p}$; $C_p=\frac{7}{2}R$;

$\mu=29\cdot10^{-3}$ кг/моль $\Rightarrow T_2\approx 193$ К. $\Delta Q=0\Rightarrow$

$\Rightarrow \frac{p_1^{\gamma-1}}{T_1^{\gamma}}=\frac{p_2^{\gamma-1}}{T_2^{\gamma}}\Rightarrow p_1=p_2\left(\frac{T_1}{T_2}\right)^{\frac{\gamma}{\gamma-1}}$; $\gamma=\frac{7}{5}\Rightarrow p_1\approx 3{,}37$ атм.

№ 6.43.

1) $V_0 \xrightarrow{T_0=const} \frac{V_0}{2}$; 2) $\frac{V_0}{2}\xrightarrow{\delta A=0} 2V_0$; Дано: a, b, C_V, T_0, V_0. ΔS – ?

$\Delta S=\Delta S_1+\Delta S_2$. Из № 6.52: $dS=\frac{C_V}{T}dT+R\frac{dV}{V-b}\Rightarrow$

$\Rightarrow \Delta S_1=R\ln\frac{V_0/2-b}{V_0-b}$; $\Delta S_2=R\ln\frac{2V_0-b}{V_0/2-b}+C_V\ln\frac{T_2}{T_0}.$

Аналогично № 6.52 происходит расширение в вакуум

$\Rightarrow U=const\Rightarrow C_VT_0-\frac{2a}{V_0}=C_VT_2-\frac{a}{2V_0}\Rightarrow\frac{T_2}{T_0}=1-\frac{3a}{2T_0V_0C_V}\Rightarrow$

$\Rightarrow\Delta S=R\ln\frac{2V_0-b}{V_0-b}+C_V\ln\left(1-\frac{3a}{2T_0V_0C_V}\right).$

Ответ: $\Delta S=R\ln\frac{2V_0-b}{V_0-b}+C_V\ln\left(1-\frac{3a}{2T_0V_0C_V}\right).$

№ 6.68.

$\underline{\alpha=0}$. $\frac{\Delta T}{\Delta p}=\frac{1}{C_p}\left(T\left(\frac{\partial V}{\partial T}\right)_p-V\right).$ $\left(\frac{\partial V}{\partial T}\right)_p\left(\frac{\partial p}{\partial V}\right)_T+\left(\frac{\partial p}{\partial T}\right)_V=0$

(при $p=const$) $\Rightarrow$ т.к. $p=\frac{RT}{V}-\frac{a}{V^2}\Rightarrow\left(\frac{\partial p}{\partial T}\right)_V=\frac{R}{V-b}$;

$\left(\frac{\partial p}{\partial V}\right)_T = -\frac{RT}{(V-b)^2} \Rightarrow \left(\frac{\partial V}{\partial T}\right)_p = \frac{V-b}{T} \Rightarrow \frac{\Delta T}{\Delta p} = \frac{-b}{C_p}$. $\Delta p < 0 \Rightarrow \Delta T > 0$ ч.т.д.

№6.69.

$b=0$. $p = \frac{RT}{V} - \frac{a}{V^2} \Rightarrow \left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{V}$; $\left(\frac{\partial p}{\partial V}\right)_T = \frac{2a}{V^3} - \frac{RT}{V^2} \Rightarrow$

$\Rightarrow$ Аналогично №6.68 из $\frac{\Delta T}{\Delta p} = \frac{1}{C_p}\left(T\left(\frac{\partial V}{\partial T}\right)_p - V\right) \Rightarrow$

$\Rightarrow \frac{\Delta T}{\Delta p} = \frac{2a}{RTC_p} \Rightarrow$ при $\Delta p < 0 \to \Delta T < 0$. ч.т.д.

№6.87.

Дано: $T_1 = 3T_{кр}$; $T_{кр} = 126\,K$; $V_{кр} = 114\,\frac{см^3}{моль}$. V_0 - ? T_2 - ?

Из сохранения энтальпии: $T_2 - T_1 = \frac{\frac{bRT_1}{V_0-b} - \frac{2a}{V_0}}{R+C_V} \Rightarrow$

$\Rightarrow T_2(V_0) = T_1 + \frac{\frac{bRT_1}{V_0-b} - \frac{2a}{V_0}}{R+C_V}$. Т.к. рассматривается максимально возможное охлаждение, $\frac{dT_2}{dV_0} = 0 \Rightarrow$

$\Rightarrow \frac{1}{R+C_V}\left(\frac{-bRT_1}{(V_0-b)^2} + \frac{2a}{V_0^2}\right) = 0 \Rightarrow \frac{bRT_1}{(V_0-b)^2} = \frac{2a}{V_0^2}$.

$V_{кр} = 3b$; $T_{кр} = \frac{8a}{27Rb} \Rightarrow a = \frac{9}{8} R V_{кр} T_{кр} \Rightarrow$

$\Rightarrow V_0^2 \cdot \frac{1}{3} V_{кр} \cdot 3RT_{кр} = 2 \cdot \frac{9}{8} R V_{кр} T_{кр} \left(V_0^2 - \frac{2}{3} V_0 V_{кр} + \frac{V_{кр}^2}{9}\right) \Rightarrow$

$\Rightarrow 4V_0^2 = 9V_0^2 - 6V_0 V_{кр} + V_{кр}^2 \Rightarrow 5\frac{V_0^2}{V_{кр}^2} - 6\frac{V_0}{V_{кр}} + 1 = 0 \Rightarrow$

$\Rightarrow \frac{V_0}{V_{кр}} = 1$ (корень $\frac{1}{5}$ не подходит, т.к. $V_0 \geq V_{кр}$) $\Rightarrow$

$\Rightarrow V_0 = V_{кр} = 114\,\frac{см^3}{моль}$. $T_2 = 3T_{кр} + \frac{\frac{V_{кр} \cdot R \cdot 3T_{кр}}{3(V_{кр} - V_{кр}/3)} - \frac{2 \cdot \frac{9}{8} R V_{кр} T_{кр}}{V_{кр}}}{R + \frac{5}{2}R} =$

$= 3T_{кр} - \frac{3T_{кр} \cdot 2}{4 \cdot 7} = \frac{39}{14} T_{кр} \Rightarrow T_2 = 351\,K$.

Ответ: $V_0 = 114\,\frac{см^3}{моль}$; $T_2 = 351\,K$.

№2.20.

Дано: $M = 1$; $p_a = 0{,}3$ атм, p_n - ? Ур-ние Бернулли для

...движения воздуха отн. ракеты: $u_a + \frac{p_a}{\rho_a} + \frac{v_a^2}{2} = u_н + \frac{p_н}{\rho_н} \Rightarrow$

7.75. $\Rightarrow \frac{i}{2}\frac{RT_a}{\mu} - \frac{i}{2}\frac{RT_н}{\mu} + \frac{R}{\mu}(T_a - T_н) = -\frac{(v_{зв}M)^2}{2}$, $v_{зв}^2 = \gamma\frac{RT_a}{\mu} \Rightarrow$

$\Rightarrow \frac{C_V}{\mu}(T_a - T_н) + \frac{R}{\mu}(T_a - T_н) = -\frac{\gamma R T_a M^2}{2\mu} \Rightarrow$

$\Rightarrow C_p(T_a - T_н) = -\frac{\gamma R M^2 T_a}{2} \Rightarrow C_p\left(1 - \frac{T_н}{T_a}\right) = \frac{-\gamma R M^2}{2}$.

Из усл. адиабатичности: $\frac{p^{\gamma-1}}{T^\gamma} = const \Rightarrow$

$\Rightarrow 1 - \left(\frac{p_н}{p_a}\right)^{\frac{\gamma-1}{\gamma}} = -\frac{\gamma R M^2}{2C_p} = -\frac{\gamma M^2(\gamma-1)}{2\gamma} \Rightarrow 1 + \frac{\gamma-1}{2}M^2 = \left(\frac{p_н}{p_a}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow$

$\Rightarrow p_н = p_a\left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}} \Rightarrow p_н \approx 0{,}6$ атм.

№ 6.73

Дано: $\Delta p_{ДТ} = \Delta p_a$; $\frac{T_{кр}}{T} = 0{,}4$; $\frac{V_{кр}}{V} = 0{,}09$. $\frac{\Delta T_{ДТ}}{\Delta T_a}$ – ?

$\frac{\Delta T_{ДТ}}{\Delta p} = \frac{T\left(\frac{\partial V}{\partial T}\right)_p - V}{C_p}$; $\frac{\Delta T_a}{\Delta p} = \frac{T}{C_p}\left(\frac{\partial V}{\partial T}\right)_p$; $\left(\frac{\partial V}{\partial T}\right)_p T - V \approx \frac{2a}{RT} - b$;

$T\left(\frac{\partial V}{\partial T}\right)_p = \frac{2a}{RT} - b + V \Rightarrow \frac{\Delta T_{ДТ}}{\Delta T_a} = \frac{\frac{2a}{RT} - b}{\frac{2a}{RT} - b + V}$.

Аналогично № 6.87, $b = \frac{V_{кр}}{3}$, $a = \frac{9}{8}RV_{кр}T_{кр} \Rightarrow$

$\Rightarrow \frac{\Delta T_{ДТ}}{\Delta T_a} = \frac{\frac{27}{4}\frac{T_{кр}}{T} - 1}{\frac{27}{4}\frac{T_{кр}}{T} - 1 + \frac{3V}{V_{кр}}} \Rightarrow \frac{\Delta T_{ДТ}}{\Delta T_a} \approx 0{,}05$.

Второе задание.

№ 19.

$v \in [0; v_0]$. $\langle v \rangle$ – ? $v_{кв}$ – ? Δv – ? $\frac{\Delta v}{\langle v \rangle}$ – ? $\int_0^{v_0} f(v)\,dv = 1 \Rightarrow$